Engineering Physics - Semiconductor Physics & Laser Optics

Course: Engineering Physics | Semester 2 | Contributor: Rohan Verma

Band theory, Hall effect, Einsteins coefficients, and optical fiber attenuation parameters.

Unit 1: Quantum Mechanics & Energy Bands

* De Broglie hypothesis, Heisenberg uncertainty, and 1D time-independent Schrodinger equation.
* Kronig-Penney model and formation of conduction and valence energy bands.
* Direct vs indirect bandgap semiconductors and Fermi-Dirac distribution function.

Unit 2: Lasers & Fiber Optics

* Population inversion, stimulated emission, and resonant cavity feedback.
* Numerical aperture NA and acceptance angle derivations for step-index fibers.
* Intermodal dispersion and Rayleigh scattering attenuation mechanisms.

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